

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

Long-horizon video generation is evaluated with whole-frame metrics that reward motion and temporal consistency. For fixed-camera nature scenes this creates an ambiguity: motion of water, fire, smoke, or rain is desirable, whereas motion of the background is an error. A system can therefore score well on motion while its scene drifts, or on consistency while its flow stagnates. We introduce SNF-Bench, an evaluation framework for long-horizon fixed-camera generation that partitions each scene into static support and dynamic flow and reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, never as one score. Each factor is validated mechanistically rather than by correlation with preference: we inject global translation, rotation, and scale drift and progressive motion attenuation at known severity into real generations, and require each factor to respond in its stated direction and to remain selective against corruptions it does not target. Auditing seven publicly released long-horizon text-conditioned generators from their released checkpoints, plus an image-conditioned deployment-sensitivity panel reported separately, we find that whole-frame motion and static-region drift induce near-opposite orderings of the same outputs, and that controlled drift which triples our static-fidelity factors moves a widely used whole-frame motion score by roughly a tenth—in the direction that rewards it. SNF-Bench measures where motion occurs and whether it persists; it does not measure physical realism.

1. Introduction

Autoregressive and streaming video diffusion models now extend generation from seconds to minute-scale horizons. Evaluation has not kept pace: small temporal errors invisible in short clips accumulate over long rollouts into background displacement, colour drift,

or motion decay, and whole-frame metrics summarise these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature scenes. In a static-camera video of a river, the water should move while the bank, rocks, and surrounding landscape stay fixed. The same separation governs waterfalls, ocean waves, rainfall, fire, smoke, and wind-driven vegetation: some regions form static support that must remain structurally fixed, while others carry dynamic flow that must persist. Whole-frame motion measures reward optical flow regardless of its origin, crediting background drift as readily as genuine fluid motion, while temporal-consistency measures can favour videos whose stability comes from suppressing the flow they should sustain. Two opposite failures—static drift and flow decay—are thereby conflated, and increasingly so at long horizons, where accumulated background motion can dominate a sequence’s apparent dynamics. Figure 1 shows both failures in released generations of a single prompt.

We introduce SNF-Bench (Static-Natural Flow), an evaluation framework built around this decomposition, evaluated at 60 and 120s horizons with shorter and longer tiers retained as diagnostics. Measurement is organised into three factors. Static fidelity captures geometric displacement and residual optical flow inside nominally stationary regions. Flow persistence reports the retention of dynamic-region motion together with its absolute compensated magnitude, so a retention ratio cannot appear strong merely because motion is uniformly weak. Drift leakage relates static-region motion and global-motion compensation to the measured dynamic-region flow.

A benchmark is only as trustworthy as its measurements, so we validate the protocol independently of any model comparison. Controlled perturbations inject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each fac-



Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench Dynamic Degree, computed under its published rule but read before the binarisation the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

tor to respond in the direction its definition states and to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators in their intended configurations. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, and never as a single score.
3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.

4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [8] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [26] extends it toward intrinsic faithfulness; Eval-Crafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialise: T2V-CompBench [19] targets compositional binding, VMBench [11] perceptually aligned motion quality, and WorldScore [1] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VMBench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond

145 short bidirectional clips by producing frames or tem-
 146 poral chunks causally, including CausVid [24], Self-
 147 Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12],
 148 Reward-Forcing [14], LongLive [22], and Causal-
 149 Forcing [27] with its few-step successor Causal-
 150 Forcing++ [25]; bidirectional systems such as LTX-
 151 Video [3] and Wan [21] provide short-horizon reference
 152 points. These systems differ substantially in temporal
 153 block size, training objective, memory and cache pol-
 154 icy, and horizon-extension mechanism [23], yet their
 155 evaluation typically relies on generic video metrics or
 156 method-specific qualitative analysis. That diversity
 157 motivates auditing each system in its released config-
 158 uration, and the evaluation gap motivates a system-
 159 agnostic question: does static support remain fixed
 160 while dynamic flow persists?

161 Image dynamics and nature-flow animation. Work
 162 on animating still images has long modelled natural
 163 motion explicitly: controllable fluid animation sepa-
 164 rates static image structure from specified fluid mo-
 165 tion [15], and Generative Image Dynamics models long-
 166 term scene motion through learned trajectory repre-
 167 sentations [10]. These works treat static support and
 168 dynamic content as distinct modelling targets; SNF-
 169 Bench elevates that distinction to an evaluation prin-
 170 ciple for long-horizon generators.

171 3. SNF-Bench Design

172 SNF-Bench evaluates long-horizon generation under a
 173 deliberately constrained setting: a fixed camera observ-
 174 ing a scene containing both stationary structure and
 175 persistent natural motion. This isolates two failure
 176 modes that are difficult to separate in unconstrained
 177 video—static drift, where nominally stationary struc-
 178 ture moves over time, and flow decay, where intended
 179 dynamics weaken or disappear. Every sequence is
 180 treated as a composition of two spatial roles, Ω_{static}
 181 for regions expected to remain geometrically station-
 182 ary and Ω_{flow} for regions expected to exhibit sustained
 183 motion. The partition defines an evaluation role, not a
 184 semantic ontology: a riverbank is static support while
 185 the river surface is flow; buildings are static support
 186 in a rainfall scene while the falling rain is flow. SNF-
 187 Bench does not require dynamics inside Ω_{flow} to follow
 188 any physical model. It measures whether motion stays
 189 localised to the appropriate regions and persists over
 190 time without contaminating static support.

191 Every system receives identical conditioning per
 192 item, and no prompt is rewritten for a particular
 193 method. Both conditioning modalities are evaluated
 194 under the same decomposition, differing only in how

static support is established: the text-conditioned set-
 ting builds it in the opening of the sequence, while the
 image-conditioned setting takes it from a shared source
 image, so divergence there is attributable to the sys-
 tem rather than to a differently imagined scene. Both
 tracks are reported in the main text at the 60 and 120 s
 horizons; all four horizons appear in the supplementary
 material.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-
 camera natural processes—channel water, ocean waves,
 precipitation, fire and smoke, lava, and wind-borne
 motion—with the category set fixed before the final
 sweep. The strata are not equally populated: chan-
 nel water accounts for 61% of the principal prompt
 set. Reported aggregates are means over prompts, not
 category macro-averages: three of the five populated
 categories hold two prompts each, so equal category
 weighting would buy balance at a cost in variance the
 sample cannot support. The supplementary material
 instead recomputes every system both ways and states
 which conclusions survive, with complete per-category
 counts.

Four temporal regimes are used: 5 s as an initial-
 isation check, 60 s as the primary reported horizon,
 120 s as a supporting tier, and 240 s as a diagnostic
 extreme. Conclusions rest on the 60 s tier, which car-
 ries 23 prompts; the 120 s tier is reported alongside it
 as a supporting check, and the 5 and 240 s tiers—an ini-
 tialisation check and a diagnostic extreme—are in the
 supplementary material. A system that cannot reach
 a horizon under its released configuration is reported
 as unsupported rather than silently modified.

3.2. Region Partition

The partition is obtained automatically and identically
 for every sequence. Mean optical-flow magnitude over
 the leading 12% of sampled frame pairs is thresholded
 by Otsu’s method [16] into a dynamic region and its
 complement; the static region is then eroded and a
 4% frame border discarded, which together form an ig-
 nored transition band so that boundary pixels—where
 the partition is least certain and flow least reliable—
 contribute to neither region.

Restricting the estimate to an early window prevents
 circularity: deriving a partition from the entire mo-
 tion field would let accumulated drift be read as evi-
 dence that the affected region is dynamic. At 60 s the
 partition is fixed from the first 7.1 s and the metrics
 evaluated over the remainder, so it reflects each sys-
 tem’s layout—unavoidable when every system authors
 its own scene—but not the temporal failure being mea-

246 sured.
247 Two limitations bound the later conclusions. The
248 partition is automatic, so a system drifting within the
249 first 12% of its rollout can influence its own partition;
250 we therefore treat partitions as measured quantities
251 and recompute every result under systematic erosion
252 and dilation of the boundary (Sec. 5). And a two-way
253 partition cannot represent layered content: rain and
254 snow transit static support, so those pixels are assigned
255 by flow magnitude alone and static-fidelity values in
256 precipitation scenes carry an unmodelled contribution.
257 Derivation, the sensitivity study, and the annotation
258 protocol we regard as the natural strengthening of this
259 component are in the supplementary material.

260 3.3. Benchmark Manifest

261 SNF-Bench is defined by a versioned manifest fixed
262 before the final sweep, recording per item: identifier
263 and category, exact prompt, target horizons, resolu-
264 tion and frame-rate metadata, and provenance. Af-
265 ter freezing, items are never removed because a sys-
266 tem performs poorly; corrections require a new bench-
267 mark version. A failure taxonomy—unsupported hori-
268 zon, out-of-memory, runtime crash, numerical corrup-
269 tion, invalid output—is likewise fixed in advance, and
270 failures are reported alongside successful generations
271 rather than silently excluded. Coverage is counted
272 from valid per-video measurements rather than from
273 the presence of a results file, so a run that terminated
274 early is reported as partial.

275 4. Drift–Motion Evaluation Protocol

276 SNF-Bench evaluates fixed-camera generation along
277 three complementary factors: Static Fidelity, Flow
278 Persistence, and Drift Leakage. No individual score is
279 sufficient: low background motion can be achieved by
280 freezing the entire video, and high foreground motion
281 can be produced by drifting the entire scene. We there-
282 fore always report static-region stability jointly with
283 both the relative persistence and absolute magnitude
284 of dynamic-region motion.

285 Let I_t denote frame t , Ω_{static} and Ω_{flow} the region
286 masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow from I_t to
287 I_{t+1} , estimated with RAFT [20]. Each sequence defines
288 an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal dura-
289 tion, $\mathcal{E}$ beginning after a warm-up so that generation
290 transients do not define the motion reference; extents
291 are fixed in the specification.

292 4.1. Static Fidelity

293 Feature-aligned Background Drift (fBD). ORB fea-
294 tures [18] extracted inside Ω_{static} from the first frame
295 are matched against each late-window frame under

robust geometric filtering; fBD ($\downarrow$) is the median
matched-point displacement, averaged over the late
window and expressed as a percentage of the frame
diagonal. It captures accumulated geometric displace-
ment of persistent structure and is intentionally insen-
sitive to fine-scale flicker that moves no structure (for-
mal definition in Supp. ??).

Normalized Background Flow (NBF). Complement-
ing accumulated drift, NBF ($\downarrow$) is the mean optical-
flow magnitude inside the static mask, averaged over
all frame pairs and normalised by frame width for res-
olution invariance and by the inter-frame interval Δt
for frame-rate invariance, giving units of 10^{-3} frame
widths per second (Supp. ??). Here Δt is the inter-
val of the sampled pairs actually used for flow esti-
mation: methods are evaluated at their native frame
rates, which differ across the roster, and the evalua-
tion pipeline subsamples each clip to a common tem-
poral rate before measurement. Expressing NBF per
unit time rather than per frame makes the quantity
independent of both the native rate and the sampling
policy, so the measure remains comparable if either
changes.

NBF is a normalised magnitude, not a ratio: di-
viding by whole-frame flow would couple background
leakage to foreground behaviour and inflate the mea-
sure for methods whose dynamic motion decays.

511 4.2. Flow Persistence

Global-motion compensation. Coherent scene dis-
placement is removed before dynamic-region motion
is evaluated. A single translation vector is insuffi-
cient: under rotation or zoom the static-region flow
field is not constant, and its component-wise median
is near zero for a centered rotation, so a translation-
only estimate would leave exactly the corruptions Sec. 5
injects uncompensated. We therefore estimate a ro-
bust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$
from static-region correspondences under RANSAC,
and subtract the displacement field it induces from the
measured flow pointwise, giving a compensated field
 $\tilde{\mathbf{u}}_t$ (Supp. ??). A similarity rather than an unrestricted
affine model is used because affine shear absorbs le-
gitimate local deformation and over-corrects. Where
correspondences are insufficient we fall back to median
translation and record the fallback per sequence.

Motion-Compensated Foreground Flow (MCFF).
MCFF ($\uparrow$) is the mean compensated flow magnitude
inside Ω_{flow} over a temporal window. We report it on
both the early and the late window—MCFF-E and

MCFF-L—so a method with uniformly negligible motion cannot appear strong through relative measures alone (Supp. ??).

Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to early MCFF, clipped at a fixed constant to prevent unstable ratios when early motion is small (Supp. ??). FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene; we report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and F_{raw} , F_{comp} the mean raw and compensated magnitude over Ω_{flow} , all in the late window, where drift has accumulated.

Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of static-region to raw dynamic-region flow magnitude in the late window, locates the sequence’s motion energy (Supp. ??). It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus the ratio of compensated to raw dynamic-region flow, measures the signed relative change in dynamic-region motion after compensation—how much apparent motion a global displacement model accounts for (Supp. ??). Reported values are clipped to $[0, 1]$ while the signed quantity is retained for validation; negative values, which arise where local flow opposes the estimated global field, are reported rather than suppressed.

Interpretation. Neither is a causal decomposition of motion into drift and non-drift components. DLR and DAR are secondary diagnostics that divide the work between them: DLR is a translational-leakage diagnostic whose rotational response is inverted, so rotational drift is carried by fBD and DAR instead, and that division is why DLR is never read alone. DAR is signed, and its negative values are real. High DLR with moderate DAR indicates static-region motion a rigid model does not account for; it does not identify what the residual is.

Semantic guardrail. The decomposition is agnostic to what moves: a sequence rendering drifting fog where a river was requested can occupy a favourable operating point. The natural guardrail is a prompt-alignment score—CLIP-based image-text similarity [6], or a learned video-quality predictor [5] at higher cost. We do not report one, and the factors should be read accordingly: they establish where motion occurs and whether it persists, not that the depicted phenomenon is the one requested. Pairing SNF-Bench with a semantic score is how it should be used in practice.

4.4. Joint Interpretation

No factor is read alone: low drift with low compensated motion indicates an over-stabilised sequence, high compensated motion with high static-region flow indicates motion contaminated by displacement, and high retention with negligible magnitude indicates persistent but trivial motion. SNF-Bench therefore reports no single scalar score, since collapsing the factors would recreate the ambiguity it exists to expose.

Implementation convention. Videos are normalised to a common evaluation resolution and subsampled to a common flow-estimation rate; no frame interpolation is performed, since synthesising frames would alter the flow field being measured. Every estimator, threshold, window extent and constant is fixed globally before the sweep and applied unchanged to every system, and the per-sequence frame rate, resolution and compensation mode are released alongside the scores.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we validate that its measurements respond to the failure modes they target. A benchmark is not justified merely because its metrics rank differently; each must change selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity, independently of the audit in Sec. 6: perturbation strengths, masks, windows, and hyperparameters are fixed before the final results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region

content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and colour temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude.

A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight judge [17] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive attenuation of dynamic-region content drives MCFF-L and FP down while leaving the static factors comparatively unchanged—the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment. Injected displacement that carries fBD to $3.2\times$ its unperturbed value moves VBench Dynamic Degree to only $1.1\times$, and in the wrong direction: a whole-frame motion score is largely insensitive to background drift, and to the extent it responds at all it improves on precisely the corruption a fixed-camera evaluation should penalise.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25% on the factor’s target family. Rotational response is reported but not gated, and the factors divide that work explicitly: DLR is a translational-leakage diagnostic whose rotational response is inverted (-0.80), while rotational drift is carried by fBD and DAR at $+0.70$ and $+1.00$. That division is why DLR is never read alone. VBench DD is a comparison row rather than a candidate factor—it is put through the identical corruptions to show what a whole-frame score does under them—so its admission cell is left blank. Twelve reference clips per severity level.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame centre: both static-fidelity factors respond more weakly than under translation ($+0.70$ for fBD, $+0.60$ for NBF, against $+1.00$ for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (-0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how deployed systems occupy the static-fidelity/flow-persistence space when used as their authors intended.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with ac-

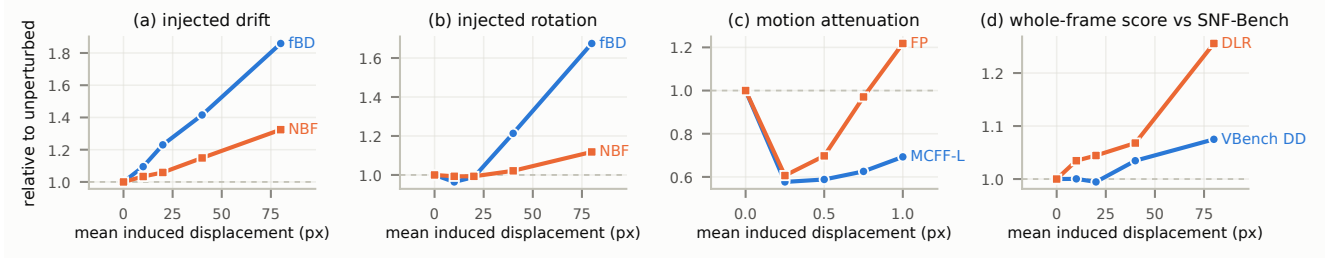


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (d) puts VBench Dynamic Degree through the identical corruption as the SNF factors. Displacement that raises fBD to $3.2\times$ its unperturbed value moves the whole-frame score by $1.1\times$ —and upward: the generic score is close to blind to where the motion originates, and what little it registers it counts in the drifting system’s favour. The perturbation is synthetic, so the comparison implicates no published method.

cessible checkpoints and reproducible inference code, frozen as a set before the sweep. Unpublished and internal systems are excluded by construction, as are systems whose outputs cannot be independently reproduced. Each is run from its released checkpoint through the recorded pipeline, with block size, denoising schedule and memory policy left as the release sets them. The scope of that claim is narrow by necessity: the run manifest records checkpoint, resolution, frame rate and horizon but not per-method sampler settings, so these are public checkpoints as executed here, not a verified reconstruction of each author’s intended configuration. The audited systems share a deployment envelope—all generate 832×480 at 16 fps and reach 240s without modification to their released temporal policy—so no comparison here is confounded by output format. We never substitute a block size, cache policy or schedule to populate a table entry; a system that cannot reach a horizon unmodified is marked unsupported.

The image-conditioned track is audited under the same protocol and the same frozen specification. Systems evaluated under a common long-horizon wrapper rather than their released configuration are labelled matched in every table, and their scores are read as deployment sensitivity rather than as the published method’s performance; we never rank a native row against a matched one.

Controls and statistics. Within each item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system, but because implementations consume randomness differently, identical seeds do not induce matched noise draws across systems; cross-system comparison therefore rests on the paired-item design. Videos are retained in native form and the normalisation of Sec. 4 applied unchanged.

Scores are computed per item and averaged over items; the sensitivity of every conclusion to category macro-averaging is reported in the supplementary material. Uncertainty is estimated by percentile bootstrap [2] over items, and we report 95% intervals. Exact ordinal ranks are avoided where uncertainty does not support them: systems whose intervals overlap substantially are not described as separated. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection.

To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and the induced orderings compared. The focus is interpretation change, not a claim that existing metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems in their released configurations; deployment-sensitivity results are reported separately in the supplementary material and never redefine a system’s native behaviour.

7.1. Native-Capability Audit

Table 2 reports the text-conditioned track at 60s, the tier whose sample supports method-level separation, and at 120s as a supporting tier. The 120s panel carries 6 prompts and is read as corroboration of direction rather than as an independent separation; it draws a different prompt set from the 60s panel, so the two panels are not compared column-by-column. The 5 and 240s tiers, an initialisation check and a diagnostic extreme respectively, are in the supplementary material with all four horizons.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	2.87 → 1.96	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	10.4 → 2.54	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	1.17 → 0.29	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	3.18 → 1.57	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	3.81 → 0.80	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	6.46 → 2.41	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	9.84 → 7.21	0.83 ± 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	0.36 → 0.52	1.06
Self-Forcing	21.2	6.80	4.85 → 0.82	0.85
Infinite-Forcing	16.1	2.60	0.50 → 0.32	0.64
Rolling-Forcing	23.8	10.6	3.33 → 0.77	0.38
Reward-Forcing	20.0	11.6	2.05 → 0.69	0.39
LongLive	22.5	9.66	5.10 → 1.19	0.81
Causal-Forcing	21.8	130	10.5 → 10.7	0.90

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are means over prompts $\pm$ half a percentile bootstrap 95% interval (10k resamples, fixed seed); panels with fewer than ten prompts report the point estimate alone, since an interval resampled from so few items is not informative. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

585 Motion persistence. Dynamic-region behaviour is
 586 read at the joint point (MCFF($\mathcal{L}$), FP), never FP
 587 alone: sustained flow needs non-trivial late motion and
 588 high retention. MCFF-E beside MCFF-L exposes the
 589 degenerate case, since negligible motion throughout
 590 cannot reach a high FP without a small MCFF-E.

591 Operating regimes. Read jointly, the static-fidelity
 592 and persistence columns separate low-drift/low-
 593 motion, high-motion/high-drift and low-drift/non-
 594 trivial-motion behaviour without collapsing the axes
 595 into a scalar score; the supplementary material plots
 596 this. The extremes are informative in opposite direc-
 597 tions. Infinite-Forcing has the lowest normalised back-
 598 ground flow, and is the only system whose interval on
 599 that factor excludes every other, while also showing the
 600 least surviving late motion; on feature-based drift its
 601 interval overlaps the next-ranked system, so it is not
 602 the least-drifting system outright. Causal-Forcing sits
 603 alone at the opposite corner, separated on background
 604 flow by a wide margin.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++	9.18	5.97	0.85 → 0.60	0.86
(2-step) [†]				
Causal-Forcing++	5.34	7.05	0.86 → 0.57	0.77
(1-step) [†]				
Causal-Forcing	21.1	36.4	2.80 → 2.69	0.89
(frame-wise) [†]				
Causal-Forcing++	9.94	6.91	1.30 → 0.78	1.02
(2-step, frame-wise)				
Self-Forcing [†]	17.3	5.52	1.50 → 0.69	0.89
CausVid [†]	19.9	3.31	0.58 → 0.41	0.81
Causal-Forcing++	10.4	5.25	0.53 → 0.52	1.01
(1-step, frame-wise)				
120s horizon ($n = 20$)				
Causal-Forcing++	9.65	8.81	0.89 → 0.82	0.93
(2-step) [†]				
Causal-Forcing++	7.26	9.39	0.84 → 0.66	0.86
(1-step) [†]				
Causal-Forcing	19.9	52.7	4.82 → 5.06	1.08
(frame-wise) [†]				
Causal-Forcing++	11.4	17.3	1.89 → 1.54	1.01
(2-step, frame-wise)				
Self-Forcing [†]	23.0	10.3	2.41 → 0.76	0.45
CausVid [†]	19.7	6.04	1.15 → 0.83	0.67
Causal-Forcing++	13.6	7.39	1.01 → 0.35	0.64
(1-step, frame-wise)				

Table 3. Image-conditioned track: native capability and deployment sensitivity. Unmarked rows are native-capability results; [†] rows are a deployment-sensitivity panel, not an audit of those systems. native systems run under their authors’ configuration, matched systems under the common long-horizon wrapper and are read as a deployment stress test rather than as the published method’s performance. [†] marks a system run under the common long-horizon wrapper rather than its released configuration; such rows are read as deployment sensitivity and are never ranked against native rows. n is stated per horizon and is common to every row of that panel. Values are means over prompts; intervals for this track are given with the complete per-horizon tables in the supplementary material. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

7.2. Drift Leakage Changes the Interpretation of Apparent Motion

The effect is systematic rather than anecdotal. Correlating each generic whole-frame metric with each SNF-Bench factor over method-level means (supplementary material), Dynamic Degree tracks static-region flow at $\rho = +0.93$, so the systems a generic metric reads as most dynamic are, closely, those whose backgrounds move most. Causal-Forcing ranks 1 of 7 on Dynamic Degree and 7 on NBF; Infinite-Forcing mirrors it, ranking 1 on static fidelity and 7 on apparent motion. Read with its magnitudes, Causal-Forcing’s DLR of 0.94 in-

617 dicates static-region motion energy almost equal to the
618 region intended to move, while its DAR of 0.40 indi-
619 cates that a rigid global model accounts for only part
620 of that motion. What the residual contains is not iden-
621 tified here—it may include non-rigid instability, in-
622 tended local motion, partition contamination or flow
623 error—and these measurements do not separate them.

624 7.3. Image-Conditioned Track: Native Capability 625 and Deployment Sensitivity

626 Table 3 is two things, kept apart. The two unmarked
627 rows are a native-capability result: systems whose re-
628 leased configuration reaches these horizons, measured
629 as shipped. The five marked rows are a deployment-
630 sensitivity panel: released checkpoints whose own con-
631 figuration does not reach these horizons, run under a
632 common long-horizon wrapper. This is not a seven-
633 system audit and is not presented as one; ranking a
634 marked row against an unmarked one would compare
635 one system’s design against another’s deployment. The
636 panel supports one descriptive observation: the same
637 checkpoint under a wrapper rather than its native con-
638 figuration yields materially different static-fidelity and
639 persistence scores, so a conclusion about a system is a
640 conclusion about that system as deployed. Why that
641 happens is outside what a benchmark establishes, and
642 we do not claim it.

643 The track supplies what the text-conditioned one
644 cannot: every system starts from the same source im-
645 age, so static support is given rather than invented and
646 divergence is attributable to the system. Bidirectional
647 reference systems reach only the 5s tier and are re-
648 ported there.

649 8. Limitations and Conclusion

650 SNF-Bench is intentionally narrow: long-horizon fixed-
651 camera generation, where static support and dynamic
652 flow separate. It measures neither physical real-
653 ism nor semantic correctness, cannot distinguish pro-
654 gression from repetition, and rests on an automatic
655 partition whose limits are stated in Sec. 3.2 and
656 bounded by the erosion–dilation study. Conclusions
657 unstable under that perturbation are never headline
658 claims.

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